Session 1.3

What Is AI? Foundations

A working definition, seventy-five years of history, why now, and what an agent is



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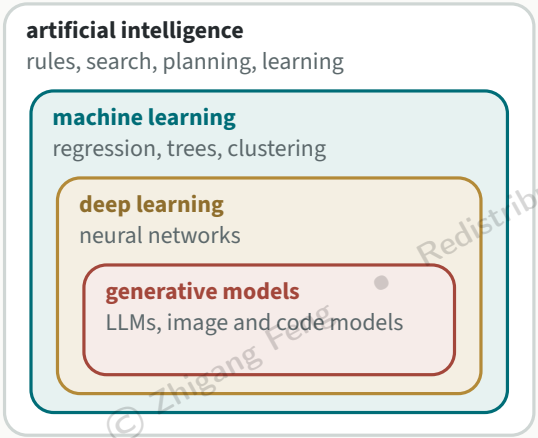
What Do We Mean by AI?

- A working definition: methods that let machines do tasks that normally need human intelligence.
- **AI is not machine learning.** Machine learning, a subfield, studies systems that improve with experience. A rule-based scheduler, or a chess program that searches moves, is AI with no learning.
- The **rational-agent** view: a system that perceives, persists, and acts for the best expected outcome. Economics has studied that object for two centuries.
- The economist's test is not "does it understand?" but what it lets you **measure, predict, identify, compute**.
- People are strong at *abduction*, the best explanation from little evidence; machines at *induction*, patterns from many examples. The boundary is moving.
- **Whether a machine thinks is philosophy. What a usable proxy or a solvable model is worth is economics.**

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Source: Russell & Norvig (2021); Miao & Feng, *AI for Economists*, §1.1; Feng, working paper (2026), §2.

The Landscape



- **AI**: systems that perform tasks associated with intelligence.
- **Machine learning**: performance improves with data or experience.
- **Deep learning**: layered representations learned by neural networks.
- **Generative models**: produce new content given a context.
- **An agent is not a fifth ring.** It is a *system* built around a model: a loop with tools, memory and permissions.

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One Question, Three Eras

Forecast next quarter's GDP growth. The question never changed. The answer did, three times.

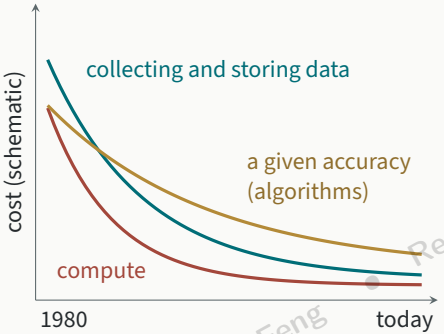
Era (approx.)	Method	Data	What is assumed
1980–2000	autoregressions, VARs	a handful of series	linear, stationary dynamics
2000–2015	factor models (Stock–Watson: 215 series), shrinkage, nowcasting	hundreds of series	a few common factors, or small coefficients
2015–	forests, boosting, neural networks	series plus text and images	little about functional form

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- **Each era added tools without retiring the old ones. Factor models and Bayesian VARs remain the benchmark that machine learning must beat.**

Source: Sims (1980); Stock & Watson (2002); Giannone, Reichlin & Small (2008); Goulet Coulombe et al. (2022).

What Drove Each Transition

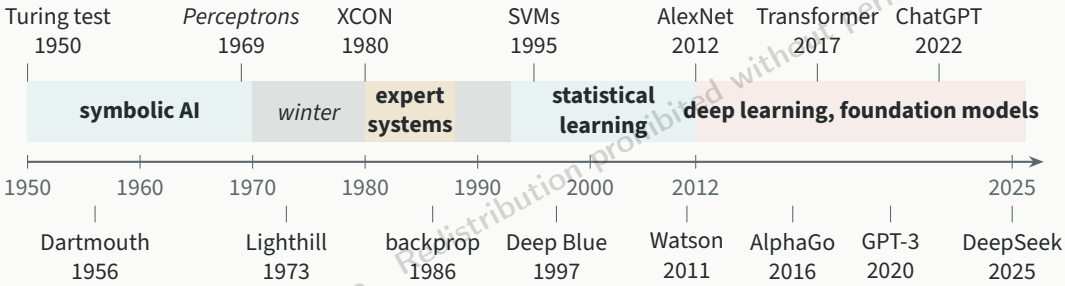


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- The ideas often existed already. They paid off when three costs fell together: **data**, **compute**, and the cost of a given accuracy.
- They are complements: one alone changes little.
- Varian: larger data sets call for new tools, such as variable selection, out-of-sample validation and trees.
- **When data and computing get cheaper, the best method changes.**

Source: Miao & Feng, *AI for Economists*, §1.2; Varian, "Big data: new tricks for econometrics", *JEP* (2014). Curves are schematic.

Seventy-Five Years in One Picture



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- Three waves, separated by two “AI winters” (grey): funding collapses when an expectations bubble bursts.
- **Each winter was also the start of the next accumulation.** The time axis stretches after 2012.

Source: Miao & Feng, *AI for Economists*, §1.3.

Wave 1: Symbolic AI and the First Winter

- **1950.** Turing replaces “can machines think?” with a test of behaviour.
- **1956.** The Dartmouth workshop names the field. Its 1955 proposal (McCarthy, Minsky, Rochester, Shannon): every feature of intelligence “can in principle be so precisely described that a machine can be made to simulate it”.
- **1965.** Herbert Simon: within twenty years machines will do any work a person can do.
- **1958–69.** Rosenblatt’s perceptron learns from examples. Minsky and Papert prove that one layer cannot learn XOR, and neural networks are shelved.
- **1973.** The Lighthill report: no part of the field had produced “the major impact that was then promised”. Britain cuts university AI funding; in the US, DARPA narrows support to mission-oriented work.
- **An expectations bubble, read as an economist: early wins extrapolated linearly, investment ahead of delivery, then a correction.**

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Source: Turing (1950); Rosenblatt (1958); Minsky & Papert (1969); Lighthill (1973); Miao & Feng, *AI for Economists*, §1.3.

Expert Systems and the Second Winter

- A new route: write a specialist's judgment down as IF–THEN rules, and let a program apply them.
- Real commercial success: DEC's XCON was saving, by 1986 and by DEC's own estimate, about $40 million a year configuring computer orders.
- Japan announced its Fifth Generation project (1981); by 1985 corporations were spending over $1 billion a year on AI, mostly in-house, and industry sales reached about $2 billion by 1988.
- The flaw was the **knowledge bottleneck**. Much of what experts know is tacit. They can do it, and cannot say how.
- It is Hayek's knowledge problem: knowledge is dispersed and hard to articulate, so centralising it in explicit rules fails once the system is complex enough.
- **The lesson of both winters: stop encoding knowledge; let the machine find patterns in examples.**

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Source: Miao & Feng, *AI for Economists*, §1.3; Hayek, "The use of knowledge in society", *AER* (1945).

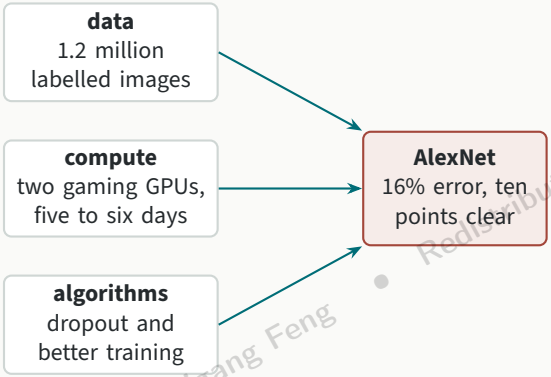
Wave 2: Statistical Learning

- **1986.** Rumelhart, Hinton and Williams make backpropagation practical: the chain rule carries the output error back through the layers, so deep networks can be trained in principle.
- In practice, compute and data were still too scarce for deep networks.
- **1990s.** Support vector machines give classification an elegant theory. LeCun's convolutional network reads handwritten digits (1998).
- The strategy: "small targets, big data". Give up general intelligence; win one measurable task at a time, such as spam, clicks, digits.
- **Deep Blue (1997) searched up to 200 million positions a second. AlphaGo (2016) learned. That contrast is the story of the next slide.**

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Source: Rumelhart, Hinton & Williams, *Nature* (1986); LeCun et al. (1998); Silver et al., *Nature* (2016).

Wave 3: The 2012 ImageNet Moment

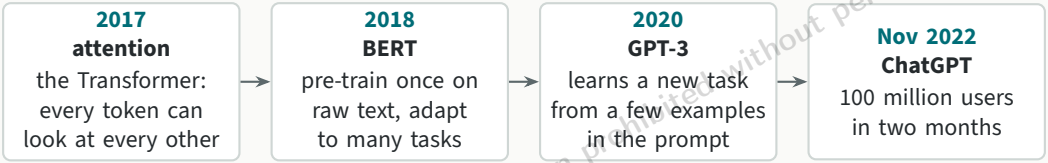


- The architecture was not new: convolutional networks date from 1989.
- New was that three complements arrived *together*. Remove any one and nothing happens.
- **Complementary innovation**: each input alone has low marginal value.
- **That is why breakthroughs are lumpy. They wait for the last complement.**

Source: Krizhevsky, Sutskever & Hinton, NIPS (2012); Miao & Feng, *AI for Economists*, §1.3.

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Transformers and Foundation Models

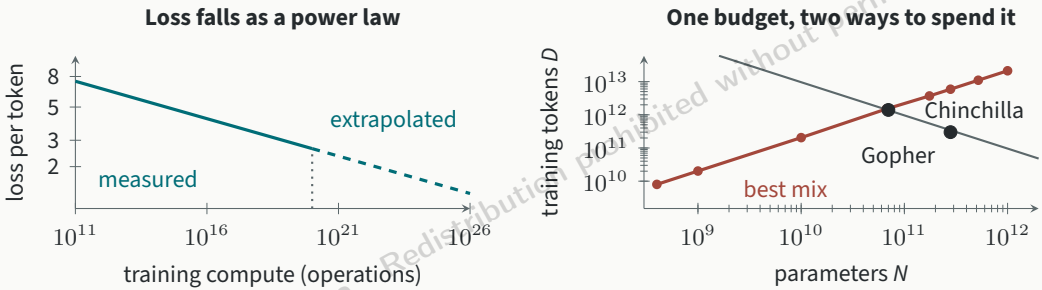


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- A **token** is a sub-word unit of text: what these models read, write and are priced in.
- **Foundation model**: one large model trained on broad data, then adapted to many downstream tasks.
- The economics changes with it: a huge fixed cost of training, a small marginal cost of use, and one model serving many markets.
- **One architecture now handles text, code, images, audio and protein sequences.**

Source: Vaswani et al. (2017); Devlin et al. (2018); Brown et al. (2020); Miao & Feng, *AI for Economists*, §1.3.

Why Now (1): Scaling Laws

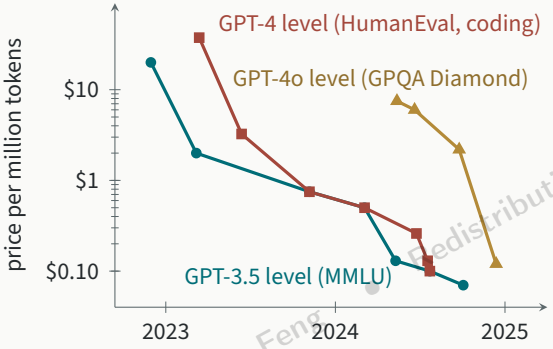


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- Ten times the compute cuts the loss by a tenth. On one budget (grey, an isocost line), Chinchilla, smaller but fed more data, beat Gopher.
- **A regularity, not a theorem. Predictable returns are what justify a trillion dollars of capex.**

Source: Kaplan et al. (2020); Hoffmann et al. (2022); Miao, Yuan & Feng, *Introduction to AI Economics*, ch. 2.

Why Now (2): Efficiency and Price



- The cheapest model matching a *fixed* capability: about 280 times cheaper at GPT-3.5 level in under two years.
- Much of it is algorithmic efficiency, not scale.
- DeepSeek-R1 (2025): o1-level reasoning; its final training runs cost a reported $6M (research excluded); GPT-4, reportedly over $100M.
- **Cheaper entry means faster diffusion.**

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Source: Stanford HAI, *AI Index Report 2025*, Fig. 1.3.22 (Epoch AI; 3:1 blended prices); DeepSeek-V3 and R1 reports; GPT-4 as reported by Altman.

2023–2026: From a Chat Box to Agent Teams



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- Alongside: *reasoning models* spend extra compute at answer time to think before they reply.
- **What changed research practice is where the loop runs and what it can touch. Each step made the work cheaper; none made checking it cheaper.**

Source: Dates checked in September 2026 against vendor announcements; Session 6 explains the protocol.

Three Ways to Learn

Paradigm	Feedback	What is learned	In economics	Session
Supervised	labelled pairs (x, y)	a mapping $x \mapsto y$	default risk, a GDP forecast	2, 5
Unsupervised	observations x only	clusters, factors, topics	grouping business cycles	5
Reinforcement	a reward along a path	a state-dependent policy	a consumption–saving rule	3

- All three pick a function from a *hypothesis space* by optimising an objective. They differ in where the feedback comes from.
- **OLS, factor models and dynamic programming are the versions you already know.**

Source: Miao & Feng, *AI for Economists*, ch. 2.

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Supervised Learning as Empirical Risk Minimisation

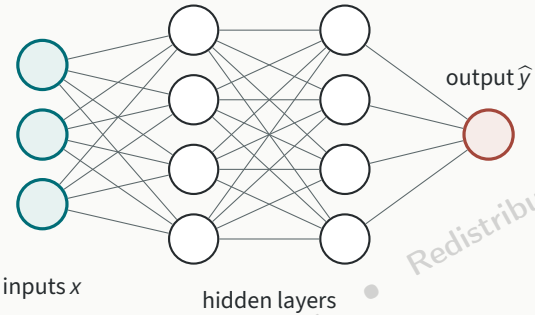
$$\widehat{f} \;=\; \arg\min_{f \in \mathcal{F}} \; \frac{1}{n} \sum_{i=1}^{n} L\big(y_i,\, f(x_i)\big) \;+\; \lambda\, R(f)$$

- Four choices define the problem: the **hypothesis space** $\mathcal{F}$, the **loss** L, the **regulariser** R with weight λ, and a **held-out rule** for judging the result.
- OLS is the special case: linear $\mathcal{F}$, squared loss, no regulariser. Lasso adds $R(f) = \|\beta\|_1$.
- A flexible $\mathcal{F}$ fits the noise in the sample. That is **overfitting**, and the held-out rule is what detects it.
- Uses in economics: nowcasts, classifying firms and households, risk and demand prediction, and nuisance functions inside causal estimators.
- **Session 2 builds on this slide: a neural network is one choice of $\mathcal{F}$.**

Source: Miao & Feng, *AI for Economists*, ch. 2.

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What a Neural Network Is



$$h^{(\ell)} = \sigma\big(W^{(\ell)} h^{(\ell-1)} + b^{(\ell)}\big)$$

- Each layer: an affine map, then a simple nonlinearity σ.
- **Width** supplies basis functions; **depth** composes them.
- The **weights** W, b are the parameters: thousands in today's lab, billions in an LLM.
- **A flexible function approximator. Session 2 builds it from one neuron.**

Source: Miao & Feng, *AI for Economists*, ch. 2; LeCun, Bengio & Hinton, *Nature* (2015).

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How a Network Learns



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- **Gradient descent**: move every weight a small step in the direction that lowers the loss.
- **Backpropagation** is the chain rule, organised so that all the derivatives cost about as much as one forward pass.
- Stochastic: each step uses a random batch, so training is a noisy walk downhill.
- **Backpropagation computes derivatives. It does not choose the loss, the data, or the claim those choices can support.**

Source: Rumelhart, Hinton & Williams (1986); Miao & Feng, *AI for Economists*, ch. 2.

From Words to Vectors to Attention

Representation	What it keeps	What it loses, or adds
Bag of words	word counts	loses order and context
Static embedding	similar words sit close together	one vector per word, whatever the sentence
Contextual embedding	meaning given the surrounding words	"rate" differs in "cut the rate" and "rate of return"
Transformer	learned attention across the whole sequence	long-range context, at a cost quadratic in length

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- **Embedding**: a map from words to vectors. Text becomes a regressor, but an *estimated* one: variables built from it carry the model's errors.
- **Attention routes information inside the text it is given. It does not retrieve truth.**

Source: Miao & Feng, *AI for Economists*, §1.5 and ch. 13; Session 5 owns the mechanics.

An LLM Predicts the Next Token

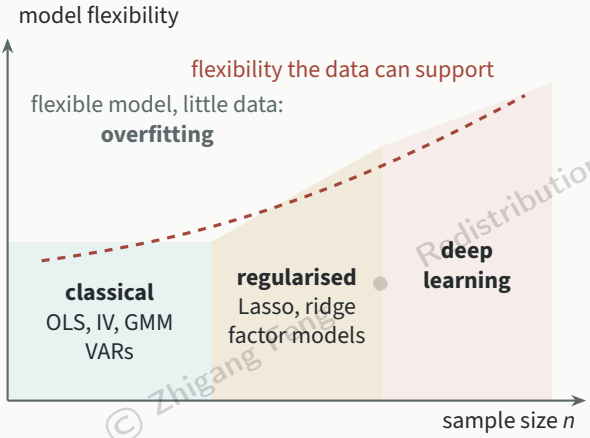
$$P_\theta(t_1, \ldots, t_T) = \prod_{i=1}^{T} P_\theta\big(t_i \mid t_{<i}\big), \qquad \mathcal{L}(\theta) = -\sum_i \log P_\theta\big(t_i \mid t_{<i}\big).$$

- A **token** is a sub-word unit of text. Training is next-token prediction; summarising, coding and dialogue come from conditioning that one objective on instructions and context.
- Output is **probabilistic**, and the objective rewards **plausibility**: sounding right and being right are different targets.
- Knowledge is bounded by the **training cut-off** and by the **context window**, the text the model can see at once.
- **Temperature** changes how the next token is sampled. It does not change what is true.
- **Session 1's lab lets you turn that dial and watch fluency and reliability come apart.**

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Source: Miao & Feng, *AI for Economists*, ch. 13.

Strong Assumptions or Flexible Data



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- Econometrics trades *assumptions* for efficiency: with about 300 quarters of post-war GDP there is little choice.
- Machine learning trades *data* for flexibility.
- "Weak assumptions" are not none. They move into the architecture, where they are harder to see.
- **Complements: theory fixes what must stay invariant; flexible methods estimate the rest.**

Source: Miao & Feng, *AI for Economists*, §1.4; Athey & Imbens, *Annual Review of Economics* (2019).

What More Data Cannot Fix

	small data	big data
the target is a prediction	a few-variable regression *limit: bias from a rigid form*	machine-learning predictors *gain: approximation bias falls with n*
the target is a causal effect	IV, RDD, difference-in-differences *limit: low power*	causal ML (double ML) *trap: unobserved confounding, now very precise*

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- Endogenous x: $\operatorname{plim}\hat{\beta} = \beta + \operatorname{Cov}(x,u)/\operatorname{Var}(x)$. More data narrows the interval, not the gap. *Identification* bias does not depend on n.
- **The Lucas critique is a distribution shift the policy itself causes: people's decision rules change with the rule, so a model fitted under the old rule fails.**

Source: Miao & Feng, *AI for Economists*, §§1.5–1.6; Lucas (1976).

What AI Is Good At, and What It Is Not

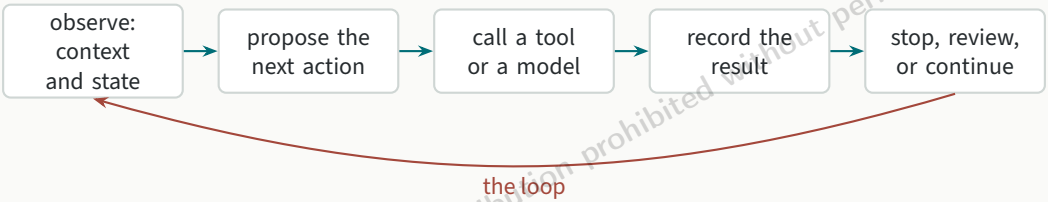
Task	Where AI is strong	The core limit	In economics
Classify, predict	many inputs, large samples	distribution shift	credit scoring
Read text	meaning, tone, topics	confident invention	central-bank communication
Read images	features, objects	needs labels	activity from satellites
Causal inference	*a helper only*	cannot settle identification	nuisance functions in double ML
Policy search	large state spaces	the reward is hard to design	simulating monetary rules

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- **AI widens what economists can measure, predict and compute. It does not change the logic of identification.**

Source: Miao & Feng, *AI for Economists*, §1.6.

From Generation to Agency

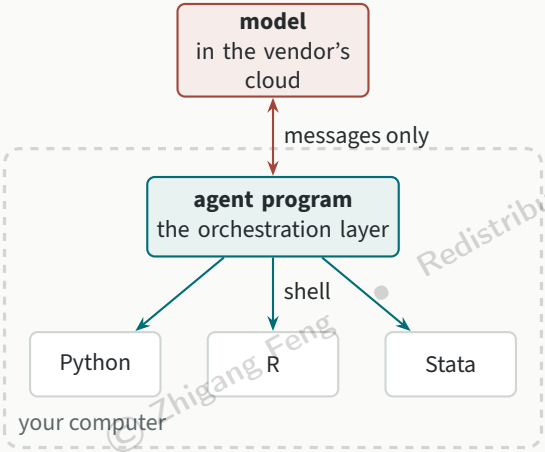


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- A language model, by itself, returns text and stops. It delivers advice.
- An **agent** is a *system*: a model, instructions, tools, memory, permissions, and a stopping rule, run in a loop. It delivers actions.
- Four marks: it uses tools, acts with autonomy, iterates, and persists across steps.
- **Change the wrapper and the behaviour changes, with the weights held fixed.**

Source: Miao & Feng, *AI for Economists*, ch. 15; Yao et al., "ReAct" (2023). Session 6 owns the architecture.

How an Agent Works on Your Machine



Source: Miao & Feng, *AI for Economists*, ch. 15.

- The agent program runs locally. It does no econometrics itself.
- It asks the model "what next?", then runs that step through the shell in the Python, R or Stata you already have.
- Files stay on your machine. Only structured messages cross to the model.
- **You do not give up Stata. You gain a layer that types commands, reads errors, and tries again.**

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Cheat Sheet

Claude Code	What it does, and why it earns a place
`/init`	writes a first standing-instructions file from the repository — the brief re-read at every start
`/plan`	enables plan mode: it explores and drafts, and changes nothing until you approve
`/model`	switches model mid-session, context kept: a derivation and a rename do not deserve the same engine
`/effort`	sets the effort level for the session — the largest quality-against-cost dial you hold
`/context`	draws context use as a grid: the invisible constraint, made visible
`/compact`	summarises the session so far — one you choose beats one that fires mid-derivation
`/rewind`	restores the conversation, or the files, to a checkpoint (Escape twice)

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Four dials: what it reads, how hard it thinks, how much it may do unasked, what it costs.

Source: Claude Code's command help, course machine, 20 September 2026. Session 6 returns to each command (Deck 6.1).

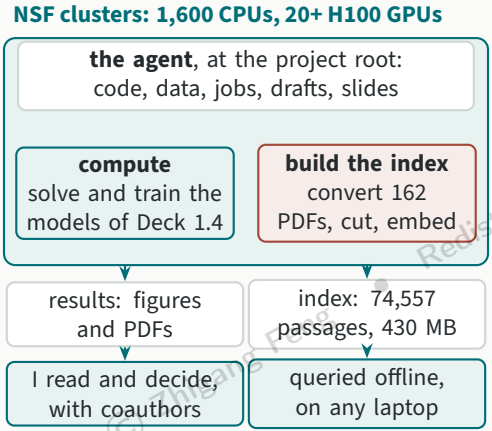
How Agents Help Economic Research

Task	What the agent does	The check you owe
Literature mapping	expands and triages a corpus	coverage; every source matches its claim
Replication audits	reruns code, compares tables	a clean run from the raw data
Data construction	finds, cleans and merges sources	spot checks against official totals
Model solution	approximates policies and values	residuals, and known special cases
Theory to code	pseudocode, tests, code variants	validation gates set before the run
Building on a literature	retrieves models by their conditions, not their wording	the conditions of each result still hold

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- The last two rows are this instructor's designs (a theory-to-code loop; condition-aware retrieval, Feng et al. 2026). Designs only; Session 6 shows them at work.
- **Delegate first the tasks whose errors are cheap to see.**

How I Work: A Personal RAG, and Everything on a Cluster



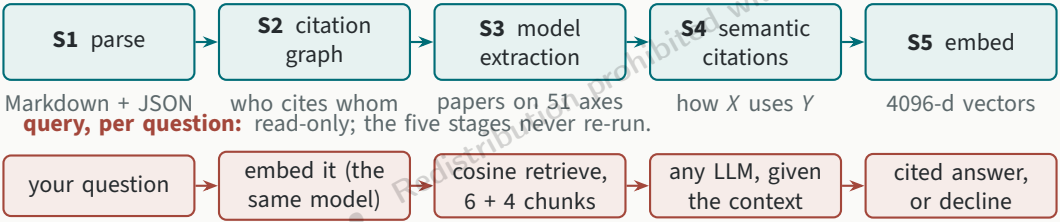
- **RAG**: the agent answers from my own library, citing the page, or says the evidence is weak.
- Build once, on a GPU: a vision model reads each PDF page (*convert*), the text is cut on its headings (*cut*), each passage becomes 1,024 numbers (*embed*).
- The same allocations solve and train Deck 1.4's models.
- **Build where the data and the GPUs are; read and decide anywhere.**

Source: The instructor's retrieval system, read 12 October 2026, and project record; allocations through NSF grants. Session 6 builds both (Decks 6.1 and 6.2).

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GRAM: A Retrieval System Built for One Literature

build once per corpus. Three LLMs in three roles: *reading* (page images to Markdown), *reasoning* (text to structured knowledge), *embedding* (text and the question to one 4096-d space).

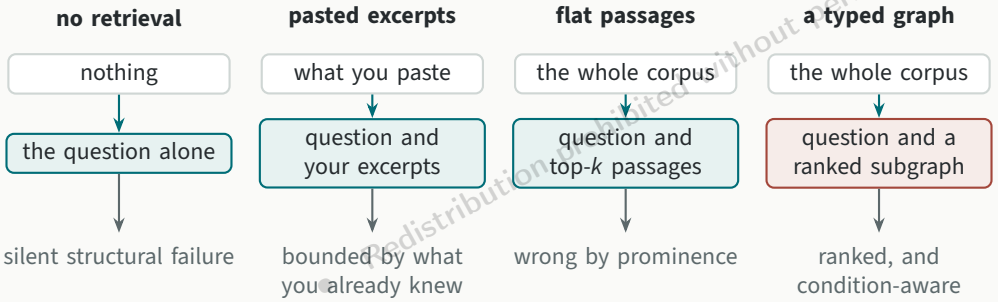


- **GRAM** (Generative Reasoning and Augmented Modeling): retrieval for one literature, quantitative macroeconomics; 133 papers in the pilot (Feng, Miao, Xie, Zhong & Zhou 2026, in progress).
- **Retrieve models by their conditions, not their wording.**

Source: Feng, Miao, Xie, Zhong & Zhou (2026), work in progress.

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Four Ways to Give a Model Context, on One Question

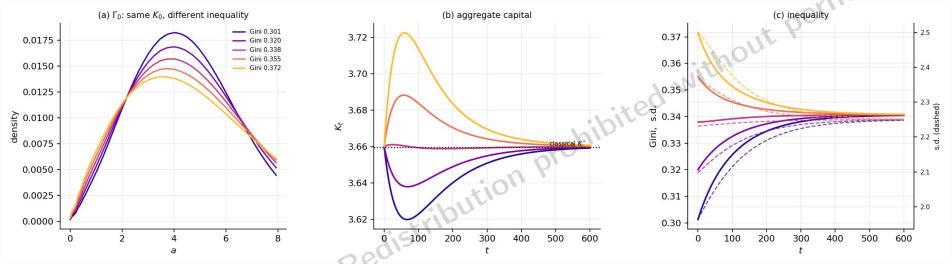


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- The generator is the same in all four. Only what reaches it changes.
- **These name each mechanism's failure mode on one kind of question. They are not a ranking of architectures.**

Source: Feng, Miao, Xie, Zhong & Zhou (2026), work in progress.

Research in Practice: Optimal Taxation With Heterogeneous Agents



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- The benchmark step of the project: five Aiyagari economies with the *same* $K_0 = 3.660$ and Gini 0.301 to 0.372 all reach the classical $K^* = 3.6595$.
- A network trained on $\Gamma_{t+1} = \Phi(\Gamma_t)$ makes convergence a result, not an assumption; Session 4 adds the tax.
- **Same K_0, yet the distributions move for hundreds of periods. K alone misses it.**

Source: Feng, Han, Sargent & Zhu (2026), work in progress. Terminal K agrees to 3.5×10^{-8}.

What Stays Human

You own

- the question, and why it matters;
- the identification strategy;
- the state variables and the equilibrium concept;
- the benchmark, chosen before the answer is known;
- the reading of a failure;
- the responsibility.

The agent speeds up

- search, reading, and first drafts;
- data cleaning and merging;
- code, tests, and variants;
- runs, logs, and figures;
- checks you have specified.

- **The division is not by difficulty. It is by who can be held to account.**

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Six Sessions: Where Each Piece Returns

Session	Object	What changes
1	Orientation	AI as tool, as object, and as an agent with incentives
2	Deep learning	Functions become trainable; equations become losses
3	Reinforcement learning	Sequential decisions are learned from interaction
4	Heterogeneous agents	Distributions and equilibrium become state variables
5	Text and LLMs	Language becomes data, measurement, and generation
6	RAG and agents	Retrieval, tools, traces, and governance in one workflow

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- Networks and training return in Session 2; learning from rewards in 3; distributions in 4; tokens and attention in 5; agents and retrieval in 6.
- **Next, Deck 1.4: the economic models these methods will solve.**